

ForgeCloud CI/CD & GitOps

GitHub Actions, Amazon ECR, declarative deployment and Argo CD

1. CI/CD Objective

GitHub Actions automates testing, image building and deployment preparation. Argo CD provides GitOps synchronization between declarative configuration and Kubernetes.

2. CI/CD Flow

```
Developer
|
Git push
|
GitHub Actions
|
Run tests
|
Build Docker image
|
Tag image
|
Push image to Amazon ECR
|
Update deployment configuration
|
Git repository
|
Argo CD
|
Kubernetes / EKS
```

3. Workflow Files

```
.github/workflows/
  test.yml
  build.yml
  deploy.yml
```

4. GitOps Model

- Desired deployment state is stored declaratively in Git.
- Argo CD observes the repository.
- Argo CD synchronizes the desired state to Kubernetes.
- Deployment changes are auditable through Git history.
- Rollback can be represented through versioned configuration.

5. Secure Pipeline Requirements

- Use GitHub Secrets for credentials.
- Never print secrets in logs.
- Use short-lived or least-privilege credentials where supported.
- Separate build and deployment permissions.

- Run tests before publishing release artifacts.
- Use immutable or traceable image tags.